

MUHAMMAD TALAL

PROFESSIONAL SUMMARY

Motivated Artificial Intelligence student with hands-on experience in AI, software development, web applications, and mobile app development. Proficient in Python, Java, React, Flutter, and database management, with strong problem-solving abilities and a passion for learning emerging technologies. Currently maintaining a CGPA of 3.29/4.00 while actively building real-world AI-driven solutions.

PROFESSIONAL SKILLS

Programming	Artificial Intelligence	Web & Development Tools
- Python, Java, C++, PHP - Algorithms & Problem Solving - Database Design (MySQL, MongoDB), Agile Collaboration	- Machine Learning - Generative AI / LLMs (RAG, Llama, LangChain) - Computer Vision - Neural Networks (CNN, RNN, ANN)	- HTML5, CSS3, JavaScript - React.js, Node.js - Flask, Django - RESTful APIs, - Flutter, React Native

EDUCATION

BS Artificial Intelligence | CGPA: 3.29/4.00

- NFC Institute of Engineering and Fertilizer Research, Faisalabad, Pakistan
- 2023- Ongoing

PROJECTS

Circuit Board Component Detection.

- Built a YOLO v5-based computer vision system to detect and classify 15+ electronic component types on circuit boards in real-time at 50 FPS with ~92% detection accuracy.
- Implemented end-to-end pipeline using PyTorch, OpenCV, and ONNX; trained on 1,000+ diverse board images with CSPDarknet backbone achieving 0.90 F1-score and edge deployment support.

BiEncoder Embedding Dimension Optimization.

- Conducted NLP research comparing 1024 vs 4096-dimensional embeddings in BiEncoder models for dense retrieval, evaluating across 2,800 queries on benchmark datasets.
- Demonstrated ~300% improvement in Recall@500 with 4096-dim embeddings using PyTorch and Hugging Face; delivered practical recommendations on dimension selection based on resource constraints.

Sentiment Classification Engine.

- Developed a Bidirectional LSTM model in TensorFlow/Keras to classify social media text as Positive, Neutral, or Negative, achieving ~85% test accuracy across 11 training epochs.
- Built a full preprocessing pipeline with NLTK covering noise removal, tokenization, stopword removal, and lemmatization; deployed a live interactive demo with real-time inference.

AI Interview Conduct System.

- Built a real-time interview proctoring platform using Flask and OpenCV with 95% detection accuracy, featuring face detection, eye tracking, head orientation monitoring, and multi-person violation alerts at 30 FPS.
- Integrated Gemini 2.5 Flash API for AI-powered question generation from PDF input, speech-to-text transcription, intelligent answer evaluation, and AI-generated content detection (0–100 scoring).

Mobile Sales Dashboard & Analytics Platform.

- Designed a 12-page Power BI business intelligence dashboard for Motorola sales data, tracking 769M in total revenue, 19K units sold, and 4K transactions with real-time KPI monitoring.
- Implemented multi-dimensional analytics using DAX, Power Query, and SQL Server covering product performance, payment method distribution (credit/debit card, cash), and temporal sales trends.

LANGUAGES

- English (Fluent)
- Urdu (Native)